PRACTICALWEEK-1

Session-1

Task1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that:

1. Accept a Linux command as input.
2. Create a child process using fork().
3. Execute the command in the child process using an appropriate exec() system call.
4. Allow the parent process to wait for the child using wait().
5. Display the Process ID (PID) of both parent and child processes.

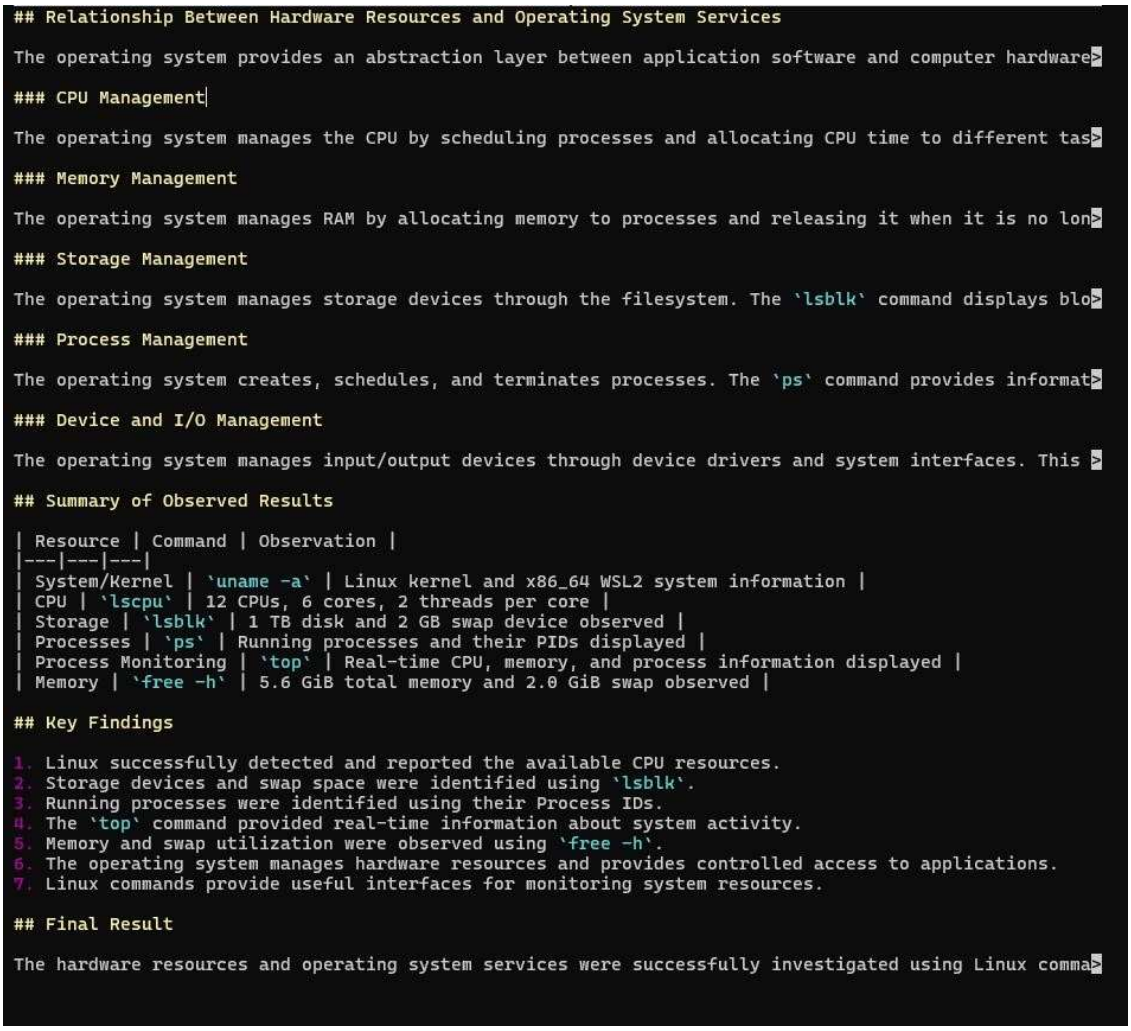
```
 GNU nano 8.7.1                                  command_executor.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char command[100];

    // Accept Linux command from the user
    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    // Create child process
    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    // Child process
    if (pid == 0)
    {
        printf("Child Process PID: %d\n", getpid());
        printf("Parent Process PID: %d\n", getppid());

        // Execute the Linux command
        execlp(command, command, (char *)NULL);

        // Executes only if execlp fails
        perror("exec failed");
        exit(1);
    }

    // Parent process
    else
    {
        printf("Parent Process PID: %d\n", getpid());
        printf("Child Process PID: %d\n", pid);

        // Wait for child process to finish
        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task1$ pwd
/home/Priyavallika/OSSP/practical/week4/task1
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task1$ ls
command_executor.c
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task1$ gcc command_executor.c -o command_executor
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task1$ ls
command_executor  command_executor.c
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task1$ ./command_executor
Enter a Linux command: ls
Parent Process PID: 659
Child Process PID: 660
Child Process PID: 660
Parent Process PID: 659
command_executor  command_executor.c
Child process completed.
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task1$
```

Observation: The program successfully accepted a Linux command as input and created a child process using fork(). The child process executed the entered ls command using exec(), while the parent process waited for the child process using wait(). The Parent PID and Child PID were displayed successfully, and the child process completed execution successfully.

Task2: Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task2$
Ghostwrite:                   Not affected
Indirect target selection:    Not affected
Itlb multihit:                Not affected
L1tf:                         Not affected
Mds:                          Not affected
Meltdown:                     Not affected
Mmio stale data:              Not affected
Old microcode:                Not affected
Reg file data sampling:       Mitigation; Clear Register File
Retbleed:                     Mitigation; Enhanced IBRS
Spec rstack overflow:         Not affected
Spec store bypass:            Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:                   Mitigation; usercopy/swapgs barriers and __user pointer
                              sanitization
Spectre v2:                   Mitigation; Enhanced / Automatic IBRS; IBPB conditional;
                              --IBRS SW sequence; BHI BHI_DIS_S
Srbds:                        Not affected
Tsa:                          Not affected
Tsx async abort:              Not affected
Vmscape:                      Not affected

Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task2$ lsblk
NAME   MAJ:MIN RM    SIZE RO TYPE MOUNTPOINTS
sda      8:0    0  356.9M  1 disk
sdb      8:16   0  159.4M  1 disk
sdc      8:32   0      2G  0 disk [SWAP]
sdd      8:48   0      1T  0 disk /mnt/wslg/distro
                             └─  /

Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task2$ ps
  PID TTY          TIME CMD
  330 pts/0    00:00:00 bash
  790 pts/0    00:00:00 ps

Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task2$ top
top -- 16:05:05 up 22 min,  1 user,  load average: 0.03, 0.02, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   5778.6 total,   5165.6 free,    528.3 used,    228.2 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   5250.3 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24124  15396  11688 S   0.0   0.3   0:00.68 systemd
      2 root      20   0    3180   2204   2072 S   0.0   0.0   0:00.00 init-systemd(Ub
      6 root      20   0    3180   2048   1948 S   0.0   0.0   0:00.00 init
     46 root      19  -1   50400  16796  15528 S   0.0   0.3   0:00.18 systemd-journal
     78 systemd+  20   0   22416  14436  11948 S   0.0   0.2   0:00.06 systemd-resolve
     81 root      20   0   35348  12304   9236 S   0.0   0.2   0:00.38 systemd-udevd
    121 root      20   0    2888   1952   1820 S   0.0   0.0   0:00.05 chronyd-starter
    143 root      20   0    4472   3148   2900 S   0.0   0.1   0:00.01 cron
    148 message+  20   0    8816   5948   4772 S   0.0   0.1   0:00.06 dbus-daemon
    157 root      20   0   44096  29888  14668 S   0.0   0.5   0:00.23 networkd-dispat
    164 root      20   0   18440   9388   8132 S   0.0   0.2   0:00.07 systemd-logind
    195 syslog    20   0  220548   5952   4648 S   0.0   0.1   0:00.08 rsyslogd
    226 root      20   0  123460  32764  18156 S   0.0   0.6   0:00.18 unattended-upgr
    233 root      20   0    5492   2744   2556 S   0.0   0.0   0:00.01 agetty
    256 _chrony   20   0   23592  11072   7128 S   0.0   0.2   0:00.08 chronyd
    271 _chrony   20   0   12096   2336   1696 S   0.0   0.0   0:00.00 chronyd
    328 root      20   0    3184   1108    988 S   0.0   0.0   0:00.00 SessionLeader
    329 root      20   0    3200   1248   1108 S   0.0   0.0   0:00.05 Relay(330)
    330 Priyavallika 20   0    6268   5536   3848 S   0.0   0.1   0:00.12 bash
    331 root      20   0    8648   5524   4680 S   0.0   0.1   0:00.01 login
    377 Priyavallika 20   0   22336  13636  11008 S   0.0   0.2   0:00.12 systemd
    379 Priyavallika 20   0   22912   4088   2088 S   0.0   0.1   0:00.00 (sd-pam)
    410 Priyavallika 20   0    6136   5528   3928 S   0.0   0.1   0:00.03 bash
    804 Priyavallika 20   0    8188   6148   4024 R   0.0   0.1   0:00.01 top

Priyavallika@LAPTOP-D2M8NVKHBH:~/OSSP/practical/week4/task2$
```



Relationship Between Hardware Resources and Operating System Services

The operating system provides an abstraction layer between application software and computer hardware>

CPU Management

The operating system manages the CPU by scheduling processes and allocating CPU time to different tas>

Memory Management

The operating system manages RAM by allocating memory to processes and releasing it when it is no lon>

Storage Management

The operating system manages storage devices through the filesystem. The `lsblk` command displays blo>

Process Management

The operating system creates, schedules, and terminates processes. The `ps` command provides informat>

Device and I/O Management

The operating system manages input/output devices through device drivers and system interfaces. This >

Summary of Observed Results

Resource	Command	Observation
System/Kernel	`uname -a`	Linux kernel and x86_64 WSL2 system information
CPU	`lscpu`	12 CPUs, 6 cores, 2 threads per core
Storage	`lsblk`	1 TB disk and 2 GB swap device observed
Processes	`ps`	Running processes and their PIDs displayed
Process Monitoring	`top`	Real-time CPU, memory, and process information displayed
Memory	`free -h`	5.6 GiB total memory and 2.0 GiB swap observed

Key Findings

1. Linux successfully detected and reported the available CPU resources.
2. Storage devices and swap space were identified using `lsblk`.
3. Running processes were identified using their Process IDs.
4. The `top` command provided real-time information about system activity.
5. Memory and swap utilization were observed using `free -h`.
6. The operating system manages hardware resources and provides controlled access to applications.
7. Linux commands provide useful interfaces for monitoring system resources.

Final Result

The hardware resources and operating system services were successfully investigated using Linux comma>

Observation: The Linux terminal commands successfully provided information about the system's hardware resources and operating system services. The uname command displayed kernel and system information, while lscpu showed CPU architecture, 12 CPUs, 6 cores, and 2 threads per core. The lsblk command displayed the available storage devices and swap space. The ps and top commands showed running processes and their resource usage. The OS abstracts CPU, memory, storage, and I/O devices by managing them through system services and providing applications with controlled interfaces instead of requiring direct access to the hardware.